

AI Tools

I explored the potential of generative AI tools (Gemini) to transform abstract concepts into functional interfaces. The objective was to create a fitness application with a "**Cute**" aesthetic based on the minimalist style of *Carrd*.

AI acts as a "concept translator," but the precision of the output depends directly on the quality of the prompt and the visual reference provided by the human.

- **UI Quality:** The tool produced clean HTML, CSS and JavaScript code. It successfully implemented a "minimalist aesthetic" which, according to Nielsen (1994), is vital so that irrelevant units of information do not compete with relevant ones [1].
- **Experience:** It was a highly iterative process. The ease of use lies in the AI's ability to handle technical syntax, allowing me to focus on visual hierarchy.

During development, specific features were integrated to enhance the User Experience (UX) through the application of heuristic principles. A "Split Workout" guide and a "Cardio Timer" were included.

The real value of AI is not just generating code, but its ability to apply established design patterns (such as error prevention) almost automatically.

- **Usability Attributes: Recognition rather than Recall** was prioritized by including dynamic exercise explanations. This reduces cognitive load, a fundamental concept in Human-Computer Interaction (HCI) to improve user efficiency [2].
- **Error Prevention:** The AI suggested input validations for empty fields, fulfilling the heuristic of helping users recover from errors without complex technical language.

After experimenting with automatic code generation, a reflection emerges regarding the evolution of our profession. Following a vision similar to Rogers' (2006) regarding human control versus proactive computing, I analyze the impact of these tools.

AI does not replace the UX designer; instead, it shifts their labor from technical execution toward strategic curation and design ethics.

- **Role Change:** The UX role will migrate from "creating from scratch" to "selecting and refining." AI can produce multiple UI iterations in seconds, but it

lacks the empathy required to understand the cultural or emotional context of the user (such as the specific appeal of a "Kawaii" aesthetic).

- **Constraints and Potential:** The potential is speed; the constraint is the lack of intrinsic ethical judgment. As Sharp, Rogers, and Preece (2019) point out, interaction design must be "user-centered," which requires a human interpretation of social needs and values that AI cannot yet replicate on its own [3].

References (ACM/IEEE Format)

- [1] J. Nielsen, "Usability inspection methods," in *Conference companion on Human factors in computing systems*, 1994, pp. 413-414.
- [2] I. S. MacKenzie, *Human-computer interaction: An empirical research perspective*. Waltham, MA: Morgan Kaufmann, 2013.
- [3] H. Sharp, Y. Rogers, and J. Preece, *Interaction Design: Beyond Human-Computer Interaction*, 5th ed. Indianapolis, IN: Wiley, 2019.